

calculation and never the actual prices to get data. This is not at all difficult but very necessary if you don't want to end up with nonsense data. The basic formula to get the log returns for one period is simply $\ln(P_t/P_{t-1})$. However, when you are dealing with daily correlation data for global futures you have another problem to take into account and that is the different closing times of the different markets. Using one-day returns may be counter-productive if you are trading futures across the globe. If the S&P makes a big move in the afternoon in New York, the Nikkei is likely to make a big move the next day and you end up with a date mismatch. The most common solution to this problem is to use a return over several days instead of just one. For instance, if you use a rolling 10-day return, $\ln(P_t/P_{t-10})$, as your basis of calculations you have smoothed this issue out and made it but a rounding error.

There are many ways to implement a risk-management strategy for a futures strategy by taking these relationships into account and this is an area where you may want to spend some time thinking things through before going live.

■ Optimisation and Its Discontents

Modern software can easily run through a few tens of thousands of iterations of our core strategy and tell us exactly which parameters would have been the best in the past few decades. It may be tempting to let your workstation go nuts and run through every possible permutation of the strategy to make sure you get the very best possible parameters. Unfortunately, this is another one of those potentially very expensive illusions.

Don't lose track of what we are really doing here. We are trading a concept, a general idea. What we need to know is whether that concept would have worked reasonably well in the past and then we need to make up our minds whether we believe that this same concept is likely to work well in the future. We need to stay objective in the evaluation process and look into all the details on what went well and what did not in the past simulated results. The optimisation process does just the opposite. It hides the potential problems and lulls you into a false sense of security.

Perhaps your optimisation process tells you that you should run the core strategy with a 57-day breakout period, a trend filter using 53 and 92 days on the long side and 41 and 113 on the short side. I did not bother to run this optimisation, and so those numbers are completely made up. Whatever